Pipeline execution snapshot: one real optimisation run

Salem · Rabi 2024 · 5 ha · balanced preference · NSGA-II mode original · fuzzy model official · official data only

1 Input

- POST /api/farm-plans {location: Salem, season: Rabi, year: 2024, total_land_ha: 5, preference: balanced, save_results: true}

2 Validation — 63.2 ms

- Request accepted by the Pydantic FarmPlanRequest model and the district / season checks.
- Rejected example → HTTP 422: Salem, Rabi, -1 ha — Input should be greater than 0
- Rejected example → HTTP 400: Salem, Summer, 5 ha — Unknown season 'Summer'.
- Rejected example → HTTP 400: Pollachi, Rabi, 5 ha — Pollachi is a taluk of Coimbatore district.

3 Official dataset loading — 668.1 ms

- 10 registered official datasets, 39,065 clean rows (report year 2024-25), each with a sha256 in dataset_registry.
- Pipeline frames built from the database: crop 985, land 76, rainfall 456, water 456 rows.
- 9 features disabled because their official dataset is missing (no estimates).

4 Preprocessing & crop filtering — 2.9 ms

- Salem · Rabi 2024 · months Oct, Nov, Dec, Jan, Feb
- Rainfall 544.8 mm vs IMD normal 341.6 mm (+59.5 %) · reservoir storage 87.44 % (Mettur) · Soil Health Card fertility index 30.57 (N Low, P Medium, K High)
- Candidates (6): maize, groundnut, jowar, ragi, red gram, green gram
- Excluded (8): cotton, cowpea, horse gram, onion, other pulses, rice, samai, varagu — e.g. cotton: no official kapas-to-lint conversion factor loaded

5 Fuzzy inference — 166.3 ms

- Model 'official': rainfall surplus, water high, soil medium → yield confidence 0.532, irrigation risk 0.12 (applied: True)

6 NSGA-II optimisation — 329.0 ms

- Population 80 × 200 generations = 16,000 evaluations · crossover 0.85, mutation 0.12 · seed 42
- 80 Pareto-optimal plans, constraint satisfaction 100% · hypervolume ratio 0.876, IGD+ 0.116, spread 0.527

7 TOPSIS ranking — inside the API call

- 50 plans ranked with the 'Balanced' profile: weights yield 0.300, production cost 0.233, irrigated area 0.233, environmental impact 0.233
- Top closeness: #1 0.6398 · #2 0.6354 · #3 0.6350 · #4 0.6266 · #5 0.6227

8 Recommendation — 515.8 ms

- Rank 1: ragi 82.5 % + green gram 17.5 % · yield 15.776 t · cost ₹880,840 · irrigated 1.244 ha-eq. · nutrients 377.253 kg
- High yield, High cost, Low water use, Low environmental impact (confidence: moderate) · saved as run salem_rabi_2024_20261112T185732Z (HTTP 200; time = whole API call)

9 Explanation — 34.3 ms

- GET /api/results/{run_id}/explanation → Recommended, confidence 60.7 %, 5 reasons, 4 risks, 19 sections
- Reasons: Best balance for the chosen preference; High yield; Low irrigation demand; Low fertiliser load; Soil supports the crops
- Confidence components: climate_soil 51.4, data_quality 100, convergence 92.7, constraints 100, historical 0, robustness 20
- Per-crop official evidence: ragi — rainfall supported, soil caution, water supported, budget info, fertiliser info, market_price unavailable, msp supported, duration info, yield caution, risks caution; green gram — rainfall supported, soil caution, water supported, budget info, fertiliser info, market_price unavailable, msp supported, duration info, yield caution, risks caution

Recorded by final_ppt/assets/scripts/run_pipeline_snapshot.py through the FastAPI app (TestClient) and the pipeline modules; times measured with time.perf_counter (fuzzy time includes building the 'official' model on first use; NSGA-II time is the run's own optimize_seconds). Raw record: final_ppt/assets/data/pipeline_run.json.